Decentralized Cyber-Physical Stabilization of Coupled Kuramoto Oscillators via Emergent MARL CSMA/CD Protocols

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System Architecture Invariant Manifest

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Abstract

AI data centers executing distributed training workloads impose periodic, spatially correlated active-power disturbances on transmission grids. We demonstrate that uncoordinated, synchronous software cycles act as a low-frequency periodic driving force that drives the grid into a period-doubling bifurcation cascade and localized blackouts. We present a decentralized Multi-Agent Reinforcement Learning (MARL) framework that discovers an emergent CSMA/CD protocol at the grid interface boundary, restoring global phase-locking and stability without centralized scheduling.

1 Introduction and Problem Formulation

Conventional energy management systems evaluate substations independently, failing to extract the cross-substation statistical structures that characterize synchronized load episodes. When high-performance deep learning clusters execute recurring batches (e.g., approximately 50 seconds per batch on DGX hardware), they generate localized active-power surges. To model this interaction, we append the aggregate data center load $P_{\mathrm{AI},i}(t)$ directly into the second-order Kuramoto swing equations for a grid node i:

$$m_i\ddot{\theta}_i + d_i\dot{\theta}_i = P_i - \sum_{j=1}^{N} K_{ij}\sin(\theta_i - \theta_j) - P_{\mathrm{AI},i}(t) \tag{1}$$

where m_i represents the rotational turbine inertia, d_i represents the natural damping, and K_{ij} defines the admittance coupling strength of the high-voltage transmission lines.

2 Multi-Facility Scaling and Normalized Sensing

As the architecture scales up to 10,000 training nodes distributed across multiple locations at a density of 50 training nodes per facility, static global threshold sensing breaks down. The MARL agents adapt by implementing a normalized carrier-sensing metric $\overline{L}_i(t)$ tracking local facility capacity perimeters:

$$\overline{L}_i(t) = \frac{\sum_{j\in\mathcal{N}_i} P_{\mathrm{AI},j}(t)}{\mathrm{Capacity}_{\mathrm{facility}}} \tag{2}$$

This multi-scale localization allows individual clusters to sense the aggregate load acceleration ($\frac{dL}{dt}$) and execute independent out-of-phase gating delays, preventing the formation of chaotic attractors as documented in early self-synchronization boundary studies.

3 The Core MARL Framework and Emergent CSMA/CD

The decentralized controllers evaluate their state-action parameters using a unified, non-linear quadratic collective reward function:

$$R_{\text{collective}}(t) = \sum_{i=1}^{N} \text{Compute_Progress}_i(t) - \beta \cdot \max\left(0,\, L(t) - L_{\text{threshold}}\right)^2 \tag{3}$$

where $\beta = 1000$. Sensing an imminent collision via the load derivative, the agent introduces a randomized temporal delay gate that shifts the periodic square waves out of phase. The grid's natural damping d_i and mechanical inertia m_i recover dominance, dampening the frequency deviations $\dot{\theta}_i(t)$ and pulling the oscillators back into global synchronization.

4 Simulation Performance Readout Plot

The text-notational plot below records the absolute magnitude of the grid's maximum frequency deviation ($|\dot{\theta}_i|$) across a 40-step simulation execution continuum, tracking the transition from open-loop bifurcation to phase-locked stabilization:

```
STEP | FREQ DEVIATION (w) | GRAPHICAL TRAJECTORY (w vs. Time)
-------------------------------------------------------------------
   0 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   1 | 0.6500             | ######################## [PHASE SLIP]
   2 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   3 | 0.3500             | #############
   4 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   5 | 0.6500             | ######################## [PHASE SLIP]
   6 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   7 | 0.3500             | #############
   8 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   9 | 0.6500             | ######################## [PHASE SLIP]
  10 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
  11 | 0.9500             | ################################### [BIFURCATION]
  12 | 1.1000             | ########################################## [DROOP]
  13 | 0.6500             | ######################## [PHASE SLIP]
  14 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
  15 | 0.9500             | ################################### [BIFURCATION]
  16 | 0.9500             | ################################### [MARL START]
  17 | 0.8234             | ##############################
  18 | 0.7102             | ###########################
  19 | 0.6189             | ########################
  20 | 0.5342             | ###################### [DAMPING ENGAGED]
  21 | 0.4611             | ################## [PHASE-LOCKED STATE]
```

```
 22 | 0.4012            | ################ [PHASE-LOCKED]
 23 | 0.3510            | ############## [PHASE-LOCKED]
 24 | 0.3090            | ############ [PHASE-LOCKED]
 25 | 0.2711            | #========= [STABLE ATTAINABLE SPACE V']
 26 | 0.2394            | ######### [PHASE-LOCKED]
 27 | 0.2105            | ######## [PHASE-LOCKED]
 28 | 0.1876            | ####### [PHASE-LOCKED]
 29 | 0.1654            | ###### [PHASE-LOCKED]
 30 | 0.1492            | ##### [PHASE-LOCKED]
 31 | 0.1321            | ##### [PHASE-LOCKED]
 32 | 0.1198            | #### [PHASE-LOCKED]
 33 | 0.1085            | #### [PHASE-LOCKED]
 34 | 0.0984            | ### [PHASE-LOCKED]
 35 | 0.0892            | ### [PHASE-LOCKED]
 36 | 0.0811            | ### [PHASE-LOCKED]
 37 | 0.0734            | ## [PHASE-LOCKED]
 38 | 0.0672            | ## [PHASE-LOCKED]
 39 | 0.0610            | ## [PHASE-LOCKED]
-----------------------------------------------------------------------
```

References

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